

Examiner reports

Q1.

There was a good number of correct force diagrams. Some candidates added one or both of the components of the tension to their force diagram. In some cases these appeared as if they were additional forces. Parts (b) and (c) were done well by most candidates who seemed to have benefitted from the printed answers. There were some candidates who did not gain full marks, because they did not show sufficient working. In part (d), many candidates obtained the correct value. Some candidates used the two printed answers as expected, if they had not been able to obtain the two values for themselves. There were some very good answers to part (e), but there were also quite a few confused ones as well, with many candidates finding it hard to give a good answer.

Q2.

Candidates found part (a) quite straightforward, although some did omit g from their calculations. Part (b) was done well by some candidates, but some of the weaker candidates used the weight of the 8kg particle rather than the tension when forming an equation to find its acceleration.

Q3.

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Q4.

In part (a), a number of candidates made assumptions about the particles instead of the pulley. Part (b) was generally done very well, but some candidates ignored the instruction to form equations of motion for each particle; these candidates were penalised. Most candidates were able to find the tension.

Q5.

As a whole candidates seemed to find this question more demanding. While some made good attempts there were a number of candidates that showed weaknesses in resolving or that did not recognise that resolving was required. In part (b), there was a number of candidates who omitted the weight from their solutions, simply summing the vertical components of the two tensions.

Q6.

Generally the candidates did well on this question. In part (a) there were a relatively small number of candidates who ignored the instruction to form an equation of motion for each particle. Some candidates used a method that produced -2.8 ms^{-2} . This approach was

acceptable for part (a), but these candidates then used 2.8 in their equation to find the tension and obtained an incorrect value. In part (c), there were many very good responses. Some candidates failed to double their answer and lost the final mark. A few candidates worked with the acceleration due to gravity, rather than the acceleration that they had calculated in part (a).

Q7.

A variety of force diagrams were seen on the candidates' scripts. Some were very good, but errors included the omission of forces and showing the friction force acting up the slope. In part (b) the candidates were clearly helped by the printed answer and there were many correct responses. In some cases the candidates did not show enough working to gain full marks. Part (c) proved to be very challenging, with only a very small number of candidates getting the correct coefficient of friction. The main problem was that the candidates did not include all of the terms that were needed in their equations of motion.

Q8.

This was an unusual question and produced many pleasing responses, showing good understanding of the law of static friction. Part (a) was generally done very well. Part (b), as expected, proved more discriminating, with a significant proportion stating 2.8 N as their answer, while some gave negative frictional forces. Part (c) produced many good solutions.

Q9.

Diagrams were mostly very good with only occasional lapses in the direction of the normal reaction force or in poor labelling of forces. Part (a) proved popular and successful for most candidates. In part (b)(i) a number failed to appreciate that the forces involved included the weight component, but many went on to be successful in part (ii).

Q10.

There were some excellent solutions to this question, which was intended to discriminate. The majority knew and attempted to apply the required principles but some were let down by poor algebraic skills in parts (a) and (b). Part (c) showed limited understanding of the equilibrium of the peg. Part (d) was mostly done well, but some failed to appreciate the need to double the distance found when considering the motion of one particle. It should be noted that in a totally algebraic question, candidates should be aware that a numerical value for g is inappropriate. However, the mark scheme treated candidates who used a numerical value sympathetically where possible.

Q11.

Many candidates were successful in this question, while others appeared to have little appreciation of the directions of forces and relevant components. Errors with signs and selection of trigonometrical function were relatively frequent.

Q12.

This question produced a variety of responses. Parts (a)(i) and (ii) were mostly answered well, although some showed all the forces acting rather than those on particle A as requested. Part (a)(iii) produced irrelevant work, as many failed to appreciate the

equilibrium situation, and produced equations of motion as if a dynamic situation existed. Part (a)(iv) proved testing, but many showed a good understanding of the forces involved. Part (b) was generally done well and accurately.

Q13.

This was a popular starter question often yielding full marks, with only occasional careless errors in arithmetic in part (a) and signs in part (b).

Q14.

In part (a) diagrams were often disappointing, with common errors including an additional force in the direction of motion and the inclusion of acceleration or velocity as a force. Parts (b) and (c) were done quite well, but a significant number of candidates selected the wrong component when finding the required forces and some failed to give their final answer to the required degree of accuracy stated in the rubric.

Q15.

Many candidates were fully successful in part (a). In part (b), however, complete equations of motion were rare and components were often inappropriately applied: either incorrect or used with inappropriate forces.

Q16.

Parts (a)(i) and (ii) were often well done, although there was some use of the 'straight string' method, and some lack of understanding of relevant forces acting on the particle on the horizontal surface.

Part (a)(iii) proved discriminating as expected, but part (b) revealed a lack of understanding of the two stages of the motion.

Q17.

The significance of the direction of the resultant force was lost on many. Again answers were often disappointing, with errors finding components and confusion with directions of forces.

Q18.

A surprising number found the static part of this question very demanding, with relatively few correct responses to parts (a)(i), (iii) and (iv). Part (b) proved more successful.

Q19.

Part (a) proved to be highly discriminating, with few fully correct solutions seen. Candidates showed little understanding of the equilibrium situation of the particles. There was confusion as to which forces acted on each particle, and also between horizontal and vertical forces. The law of friction was known and used by some candidates, but it was rare to see competent handling of inequalities.

Part (b) was often started well, although the description of the surface as 'smooth' was not always appreciated by candidates. Some of the algebraic work was unconvincing, but candidates who were able to write down correct equations for the particles usually

completed their solution correctly.

Q20.

Many candidates attempted this question satisfactorily, realising that the sum of the components of the three given forces was zero parallel to Ox , but non-zero parallel to Oy . Resulting equations were often fully correct and accurate arithmetical work followed. Some errors in selecting the correct component or sign were seen, but the most common mistake was failure to find the magnitude of the resultant and thereby complete part (b).

Q21.

A surprising number of candidates failed to score both marks for the forces diagram in part (a), with the most common errors being the omission of the friction force or stating the normal reaction force to be equal to the weight. Parts (b) and (c) were done well, and many were successful in part (d), although some omitted a force from their calculations.

Q22.

This question was the least popular and successful on the paper. Many candidates showed a lack of understanding of the directions in which forces act, although there were fewer instances of including inappropriate forces than in the past. Part (a)(i) was moderately successful but with many sign errors. Part (a)(ii) was not done well, with few candidates able to write down an equation that contained all the relevant forces. Part (b) proved a further test of understanding, this time testing Newton's first Law, of which a significant number of candidates lacked knowledge.

Q23.

Diagrams in part (a) were often good, although there was a tendency to include the acceleration of the moving object on force diagrams. Part (b) was generally well done but there was some confusion with signs in part (c). Most completed part (d) but sometimes lost a mark through failing to give their answer to three significant figures as demanded in the rubric for the paper.

Q24.

Parts (a) and (b) of this question proved very successful. Some candidates had difficulty in appreciating the nature of the motion after the string broke and chose an incorrect acceleration in part (c). Graphs in part (d) were generally good.

Q25.

The candidates were more successful with this question than had been expected.

Part (a) was quite well answered. A reasonable number of candidates formed the two equations of motion correctly and then went on to obtain the required value. Some candidates did not resolve the weight of the particle on the slope. A small number of candidates did not form an equation of motion for each particle.

Those candidates who realised that the acceleration was zero did well in part (b). There were quite a few candidates who tried to use the acceleration from part (a) when tackling part (b).

Q26.

Part (a) was generally well answered, although some candidates did make errors when resolving.

Part (b) was found to be much more difficult. The majority of the candidates did not realise that they needed to include a component of the weight and the friction force acting in the same direction. Also a number of candidates did not seem to realise that the acceleration was directed down the slope.

Q27.

This question was done very well. The answer that was given did seem to have helped some candidates. In part (c), a few candidates stated that the tension was zero, even though they had done both parts (a) and (b) correctly.

Q28.

This question was generally done well by the majority of candidates, but some experienced problems. In part (b), there were a few candidates who ignored the instruction to form an equation of motion for each particle. A few others made errors in one equation. Part (c) was usually done correctly if the candidates had two equations of motion in part (b). If they did not have these equations, there were often errors in this part. Part (d) was done well using the acceleration given in the question.

Q29.

Parts (a) and (b) were generally done well. In part (b), the printed answer helped many candidates, but some could not put together a convincing argument. Part (c) was more challenging. Some candidates did resolve horizontally and apply the friction law, but then got into difficulties solving the resulting equation. Other candidates did not take both of the steps needed to produce an equation for T and made very little progress. There was also some evidence of candidates using $\sin 30^\circ$ instead of $\cos 30^\circ$ when resolving horizontally.

Q30.

The force diagram asked for in this question was drawn fairly well, but the usual errors of forces in the direction of motion and missing forces were present on some scripts. Parts (b) and (c) were done fairly well and it appeared that many 'candidates' had been helped by the printed answer in part (b). There were some candidates who did not realise that they should consider the tension in the rope when finding the normal reaction. Part (d) was found to be very much more demanding, with relatively few correct responses.

Candidate typically omitted one term from their equation or failed to solve correctly for T .

Q31.

The candidates found this question a little harder than the previous two questions. There were a number of poor force diagrams, some showing a force in the direction of motion. The errors made by candidates were mainly due to incorrect resolving, but some of these were corrected to produce the answers given in the question. A small number stated that the reaction force was 49 N.

Q32.

There were several good solutions to part (a), but some candidates ignored the instruction to use two equations of motion. Some candidates assumed that A would move up the slope and then ran into difficulties with the negative signs that were involved. Many candidates found part (b) difficult. It was very common for them to try to use the acceleration from part (a) instead of setting the acceleration equal to zero. Interestingly, some candidates who made little or no progress with part (a) did complete part (b) successfully.

Q33.

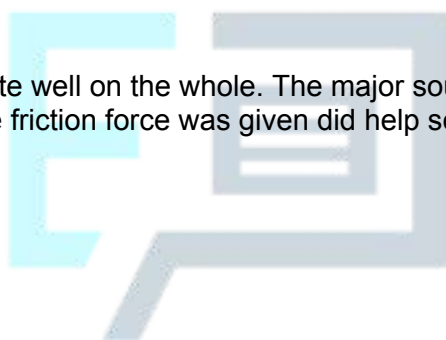
In this question, there were problems with the force diagram and resolving correctly. Many candidates were able to find the reaction force correctly, but there were more problems in part (d).

Q34.

This question was generally found to be difficult and many of the weaker candidates did not score many marks on this question. Many candidates produced the answer given in the question, but did not have satisfactory working to support their answer. Those who completed part (a)(i) correctly often went on to do well with part (a)(ii). Part (b) was found to be very difficult, with very few convincing explanations.

Q35.

This question was done quite well on the whole. The major source of errors was resolving incorrectly. The fact that the friction force was given did help some candidates to work towards the correct answer.



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